Problem 2

Problem 2

Problem 1 *Suppose we know that function f is positive and increasing, and the function g is negative and decreasing. Determine whether the following functions are increasing or decreasing.*

(a) *The product function: $f(x)g(x)$.*

Explanation. The derivative of $f(x)g(x)$ is $f'(x)g(x) + f(x)g'(x)$. Since f is increasing and g is decreasing, $f'(x)$ is positive and $g'(x)$ is negative in the interval we are considering. That means $f'(x)g(x)$ is the product of a positive and a negative number, so it is negative. Similary, $f(x)g'(x)$ is the product of a positive number and a negative number, so it is also negative. The result is that $\frac{d}{dx}\left(f(x)g(x)\right)$ is negative, so $f(x)g(x)$ is decreasing in the interval.

(b) *The composite function: $f(g(x))$.*

Explanation. The derivative of $f(g(x))$ is $f'(g(x))g'(x)$. Since f is increasing and g is decreasing, $f'(x)$ is positive and $g'(x)$ is negative in the interval we are considering. That means $f'(g(x))g'(x)$ is the product of a positive and a negative number, so it is negative. The result is that $\frac{d}{dx}\left(f(g(x))\right)$ is negative, so $f(x)g(x)$ is decreasing in the interval.